ACCOUNTS of chemical research

Article

Cite This: Acc. Chem. Res. 2019, 52, 1301−1312

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# Allenation of Terminal Alkynes with Aldehydes and Ketones

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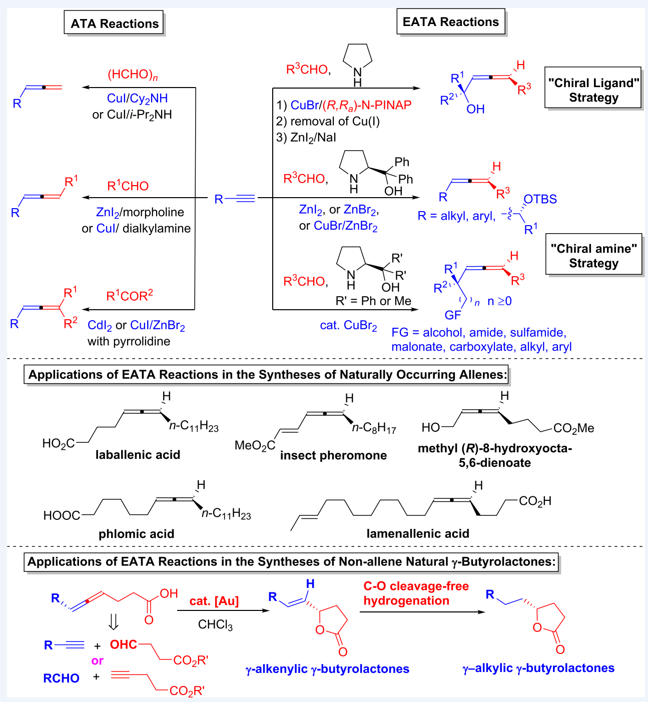
**CONSPECTUS:** So far, over 150 natural products and pharmaceuticals containing an allene moiety have been identified. During the last two decades, allenes have also been demonstrated as synthetically versatile starting materials in organic synthesis. In comparison to alkenes and alkynes, allenes are unique unsaturated hydrocarbons due to their axial chirality, which could be transformed to central chirality via chirality transfer to provide an irreplaceable entry to chiral molecules. Thus, methods for allene synthesis from readily available chemicals are of great interest.




In 1979, Crabbé et al. reported the first CuBr-mediated allenation of terminal alkynes (ATA) reaction to form monosubstituted allenes from 1-alkynes and paraformaldehyde in the presence of diisopropylamine. During the following 30 years, the ATA reactions were limited to paraformaldehyde. This Account describes our efforts toward the development of ATA reactions in the last ten years. First, we improved the yields and scope greatly for the synthesis of monosubstituted allenes by modifying the original Crabbé recipe. Next we developed the $ZnI_2$-promoted or CuI-catalyzed ATA reactions for the synthesis of 1,3-disubstituted allenes from terminal alkyne and normal aldehydes. Furthermore, we first realized the $CdI_2$-promoted ATA reaction of ketones with pyrrolidine as the matched amine for the preparation of trisubstituted allenes. Due to the toxicity of $CdI_2$, we also developed two alternative approaches utilizing $CuBr/ZnI_2$ or $CuI/ZnBr_2/Ti(OEt)_4$. The asymmetric version of ATA reactions for the synthesis of optically active 1,3-disubstituted allenes has also been achieved in this group with two strategies. One is called "chiral ligand" strategy, using terminal alkynes, aldehydes, and nonchiral amine with the assistance of a proper chiral ligand. The other is the "chiral amine" strategy, applying terminal alkynes, aldehydes, and chiral amines such as (*S*)- or (*R*)-$\alpha,\alpha$-diphenylprolinol or (*S*)- or (*R*)-$\alpha,\alpha$-dimethylprolinol. Optically active 1,3-disubstituted allenes containing different synthetically useful functionalities such as alcohol, amide, sulfamide, malonate, carboxylate, and carbohydrate units could be prepared without protection with the newly developed $CuBr_2$-catalyzed chiral amine strategy. Recently, we have applied these enantioselective allenation of terminal alkyne (EATA) reactions to the syntheses of some natural allenes such as laballenic acid, insect pheromone, methyl (*R*)-8-hydroxyocta-5,6-dienoate, phlomic acid, and lamenallenic acid, as well as some non-allene natural $\gamma$-butyrolactones such as xestospongienes (E, F, G, and H), (*R*)-4-tetradecalactone, (*S*)-4-tetradecalactone, (*R*)-$\gamma$-palmitolactone, and (*R*)-4-decalactone.

## 1. INTRODUCTION

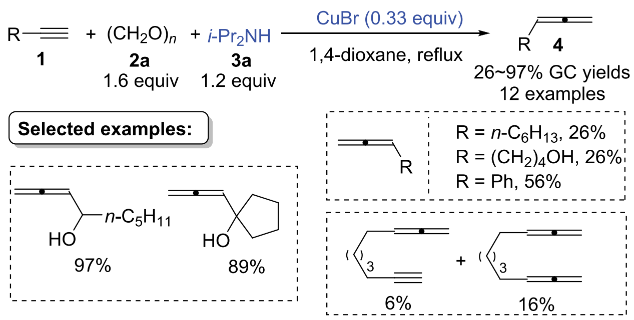
Allenes have been proven to be very powerful chemicals in organic transformations such as carbometalation, nucleometalation, electrophilic addition, nucleophilic addition, and radical addition.[1] In addition, axially chiral allenes owing to the two perpendicular C═C bonds provide an irreplaceable opportunity to obtain optically active molecules via chirality transfer strategies.[2] The ever-growing synthetic potentials of allenes in organic synthesis combined with their existence in a growing list of natural products and active pharmaceutical ingredients have stimulated chemists to make intense efforts to develop straightforward methods for their synthesis,[3] especially for the preparation of optically active allenes.[4]

In 1979, Crabbé et al. reported the first CuBr-mediated allenation of terminal alkynes (ATA) with paraformaldehyde to prepare monosubstituted allenes in the presence of diisopropylamine.[5] It should be noted that the $\alpha$-hydroxy

Received: January 11, 2019
Published: April 15, 2019

substitution significantly increased the reactivity of the alkynes, while phenyl substitution and long carbon-chain alkyl substitutions led to lower yields. Moreover, the reaction of nona-1,8-diyne only afforded monosubstituted allene product in 6% yield and bisallene product in 16% yield (Scheme 1).

**Scheme 1. Original ATA Reaction by Crabbé**




During the following 30 years, this reaction was still limited to paraformaldehyde.[3] This Account will summarize our recent efforts toward the development of the ATA reactions since 2009. First, we improved the yields and scope for the synthesis of terminal allenes by modifying the Cu promoter and the secondary amine. Then we developed ATA reactions of terminal alkynes with normal aldehydes or ketones to construct 1,3-disubstituted or trisubstituted allenes. Moreover, the enantioselective allenation of terminal alkynes (EATA) with aldehydes has also been achieved.

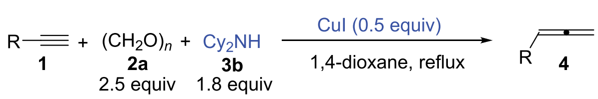
## 2. ALLENATION OF TERMINAL ALKYNES (ATA): THE RACEMIC VERSION

### 2.1. Synthesis of Monosubstituted Allenes via ATA Reactions with Paraformaldehyde

Our attempt started with developing a modified one-pot procedure for converting terminal alkynes to monosubstituted allenes. The reaction of 1-alkynes with $Cy_2NH$ (1.8 equiv) and paraformaldehyde (2.5 equiv) mediated by CuI (0.5 equiv) in refluxing dioxane may produce monosubstituted allenes in much higher yields (over 20% for most cases) than the original Crabbé's protocol. Many functional groups, such as mesylate, hydroxyl group, ether, and amide, may be tolerated (Scheme 2).[6]

This modified method was then successfully applied to the synthesis of a series of terminal *N*-allenyl amides from *N*-propargyl amides in good to excellent yields (Scheme 3).[7] Again, the yields with our protocol were much higher than that with Crabbé's protocol as reported by Hashmi.[8] These *N*-allenyl amides have shown their great synthetic potential toward oxazoline derivatives in our recent reports.[7,9]

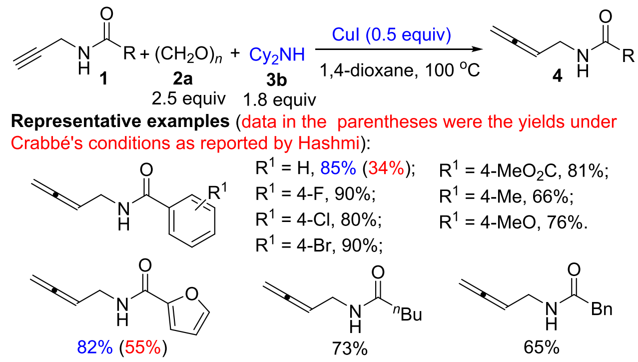
**Scheme 2. Synthesis of Monosubstituted Allenes via CuI promoted ATA Reactions with $Cy_2NH$**




**Representative examples** (data in the parentheses were the yields under Crabbé's conditions):

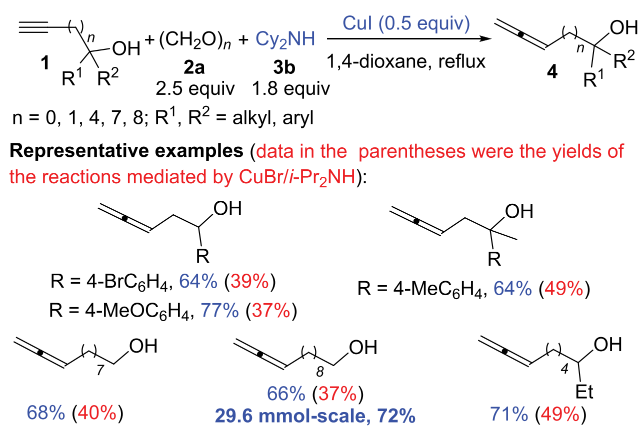
R = $n$-$C_8H_{17}$, 66% (42%);
R = $n$-$C_{10}H_{21}$, 60% (36%);
R = $p$-$PhC_6H_4$, 70% (59%);
R = $MsO(CH_2)_7$, 53% (30%);
R = $PhCH_2OCH_2$, 81% (53%);
R = $HOCH_2C(Me)_2NTsCH_2$, 42% (27%).
R = $p$-$EtC_6H_4CH(OH)CH_2$, 74% (40%);
R = $p$-$ClC_6H_4CH(OH)CH_2$, 72% (32%);
R = $o$-$ClC_6H_4CH(OH)CH_2$, 52% (33%);
R = $p$-$BrC_6H_4CH(OH)CH_2$, 65% (35%);
R = $p$-$O_2NC_6H_4CH_2OCH_2$, 98% (59%);

**Scheme 3. Synthesis of Terminal *N*-Allenyl Amides via CuI-Promoted ATA Reactions with $Cy_2NH$**




As a type of synthetically attractive functionalized allenes, allenols are of great importance in organic synthesis due to the combination of the reactive allene moiety and the hydroxy functionality.[10] With the modified $CuI/Cy_2NH$ method, $\alpha$-allenols, $\beta$-allenols, and allenols bearing a long carbon chain between the allene unit and the hydroxyl group could all be prepared in moderate to good yields. Compared with the traditional conditions of CuBr/$i$-$Pr_2NH$, the new combination of $CuI/Cy_2NH$ enjoys apparent advantages especially in the preparation of terminal allenols with long carbon chains (Scheme 4).[11]

**Scheme 4. Synthesis of Monosubstituted Terminal Allenols via CuI Promoted ATA Reactions with $Cy_2NH$**

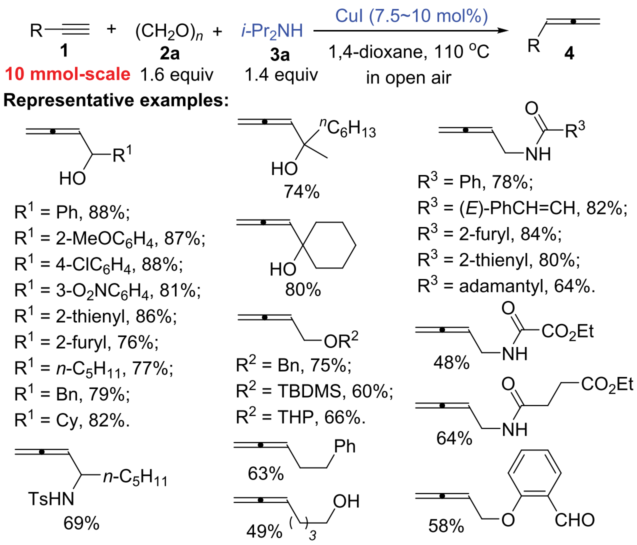



An efficient and facile protocol for the 10 mmol scale synthesis of functionalized monosubstituted allenes by using CuI (7.5−10 mol %), paraformaldehyde (1.6 equiv), and diisopropylamine (1.4 equiv) in 1,4-dioxane at 110 °C was also demonstrated. It performed well in the synthesis of protected or unprotected $\alpha$-allenols and $\alpha$-allenylamides. Nonfunctionalized 1-alkynes were also compatible with this method, although it was still somewhat lower-yielding as compared to our modified protocol (Scheme 5).[12]

Buta-2,3-dien-1-ol is the simplest $\alpha$-allenol but a useful organic building block.[13] Traditional approaches often require a multistep manipulation from but-2-yn-1,4-diol and dangerous $LiAlH_4$ as the reducing reagent.[13] Recently, after extensive screening and optimization of the workup procedure, we

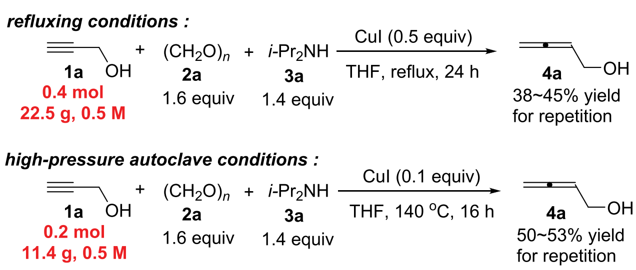

DOI: 10.1021/acs.accounts.9b00023
Acc. Chem. Res. 2019, 52, 1301−1312

**Scheme 5. Synthesis of Monosubstituted Allenes via CuI Catalyzed ATA Reactions with $i$-$Pr_2NH$**



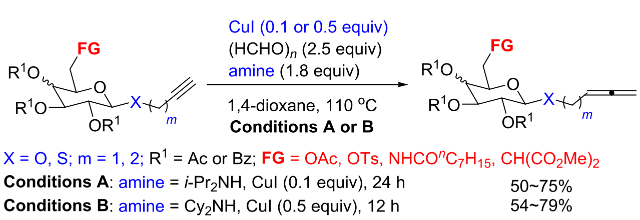

developed two practical large-scale procedures for the synthesis of buta-2,3-dien-1-ol with THF as the optimal solvent: (1) under refluxing conditions, 0.4 mol of propargyl alcohol proceeded smoothly to afford 38−45% yield of the product by using 0.5 equiv of CuI; (2) under high-pressure autoclave conditions, 10 mol % of CuI was enough to obtain a better result (Scheme 6).[12,14]

**Scheme 6. Synthesis of Buta-2,3-dien-1-ol via CuI Promoted ATA Reaction with $i$-$Pr_2NH$**




Additionally, we had also successfully demonstrated two improved CuI-mediated ATA reactions to synthesize carbohydrate-based terminal allenes in moderate to good yields, tolerating functional groups such as ether, ester, amide, and malonate units. Allenyl $\beta$-D-glucopyranosides, $\beta$-D-galactopyranoside, thioglycosides, disaccharides, and even pentakis-allenyl $\beta$-D-glucopyranoside could all be synthesized by applying these two methods (Scheme 7).[15]

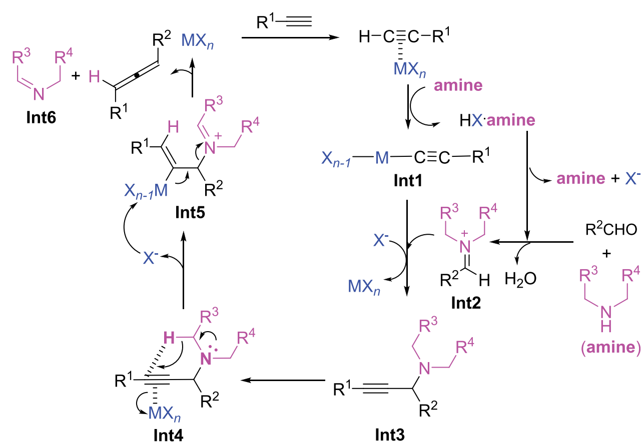
**Scheme 7. Synthesis of Carbohydrate-Based Terminal Allenes**




## 2.2. Synthesis of 1,3-Disubstituted Allenes via ATA Reactions with Normal Aldehydes

Encouraged by the improvements of the ATA reaction with paraformaldehyde, we then turned our attention to the one-pot synthesis of 1,3-disubstituted allenes via the ATA reaction with normal aldehydes. In 2008 and 2010, Wong and Che reported their pioneering works on the Au(I)- or Ag(I)-mediated synthesis of 1,3-disubstituted allenes from propargylic amines.[16] We assumed that a proper combination of metal salt ($MX_n$) and dialkylamine would be the key for the direct ATA reaction with normal aldehydes. The reaction of $MX_n$ with terminal alkyne generates the 1-alkynyl metal species, **Int1**, which reacts with the iminium ion **Int2** formed in situ from the aldehyde and dialkylamine to produce the corresponding propargylic amine **Int3** and regenerate $MX_n$. The C≡C bond in the propargylic amine then coordinates to $MX_n$ to give complex **Int4**, which undergoes a 1,5-hydride transfer and $\beta$-N elimination to afford the allene product and the imine byproduct **Int6** (Scheme 8). Overall, this is a redox reaction, requiring the presence of a secondary amine, which is oxidized to imine during the process. For further supporting mechanistic studies, see also Scheme 23.

**Scheme 8. Proposed Mechanism of ATA Reaction with Aldehydes**

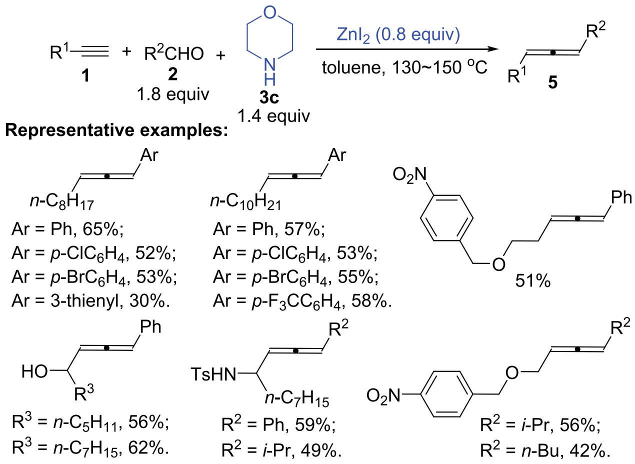



After some trial and error, we were pleased to identify the combination of $ZnI_2$ with morpholine for such a purpose. The reactions of terminal alkyl alkynes with aromatic aldehydes bearing electron-deficient groups proceeded well, producing the corresponding 1-aryl-3-alkylallenes including synthetically attractive 1,3-disubstituted $\alpha$-allenols and $\alpha$-allenyl sulfamides in moderate yields. In comparison, the reactions with heteroaryl aldehydes such as 3-thienyl carbaldehyde and aliphatic aldehydes afforded allenes in relatively lower yields (Scheme 9).[17] According to the mechanism proposed in Scheme 8, the reaction should be catalytic in $MX_n$; however, 0.8 equiv of $ZnI_2$ was required. We reasoned that the in situ generated water and the coordination of $ZnI_2$ with the in situ formed imine in the last step would be the major challenge for a catalytic reaction.

In order to develop a practical and catalytic version of such an ATA reaction with aldehydes,[18] we conducted the reaction of 1-phenyl-2-propyn-1-ol and $n$-butanal by using CuI (10 mol %) as the catalyst and found that the structure of the amine[19] is very important. The reaction using $n$-$Bu_2NH$ gave the highest yield of allenol **6a** (Scheme 10A).[20] We also noticed that if the

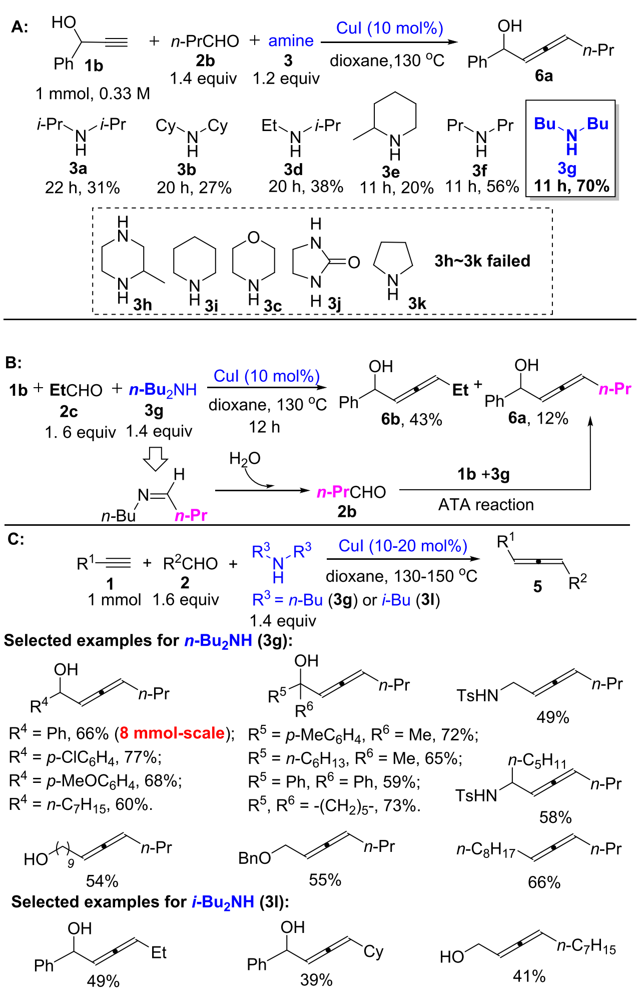

DOI: 10.1021/acs.accounts.9b00023
Acc. Chem. Res. 2019, 52, 1301−1312

**Scheme 9. Synthesis of 1,3-Disubstituted Allenes via $ZnI_2$ Promoted ATA Reactions from Terminal Alkynes, Aldehyde, and Morpholine**




dialkylamine used does not match with the aldehyde, a mixture of allenes **6a** and **6b** was produced, which was caused by the formation of butanal **2b** from the hydrolysis of the in situ generated imine (Scheme 10B).[20] Further screening on the

**Scheme 10. Synthesis of 1,3-Disubstituted Allenes via CuI Catalyzed ATA Reactions from Terminal Alkynes, Aldehyde, and Dibutylamine or Diisobutylamine**

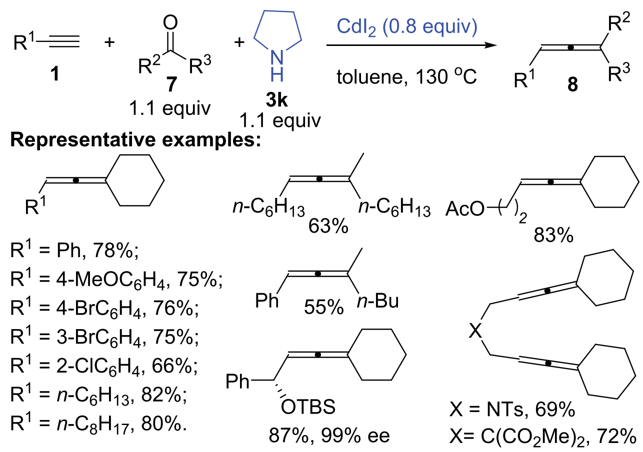



amine led to the observation that *i*-$Bu_2NH$ may be used as a common amine without contamination of the allenol derived from *i*-$Bu_2NH$, although the yields are slightly lower. Compared with the $ZnI_2$-promoted version of ATA reaction, this new catalytic protocol performed well in the preparation of alkyl-substituted 1,3-disubstituted allenes (Scheme 10C).[20] Unfortunately, terminal aryl alkynes were still not compatible.

### 2.3. Synthesis of Trisubstituted Allenes via ATA Reactions with Ketones

The much lower reactivity of ketones made it a great challenge to identify a more efficient combination of $MX_n$ and amine. Based on the previous experiences, we investigated the inorganic salts of group 11 metals (CuI, AgI, AuI) and group 12 metals ($ZnI_2$, $CdI_2$, $HgCl_2$). Fortunately, the ATA reaction with ketones succeeded under the promotion of $CdI_2$ (0.8 equiv) with pyrrolidine as the matched amine. With this protocol, a series of different trisubstituted allenes **8**, including 1,5-bisallenes and protected chiral allenols, which are especially important in organic synthesis, have been successfully prepared (Scheme 11).[21]

**Scheme 11. Synthesis of Trisubstituted Allenes via $CdI_2$ Promoted ATA Reactions from Terminal Alkynes, Ketones, and Pyrrolidine**

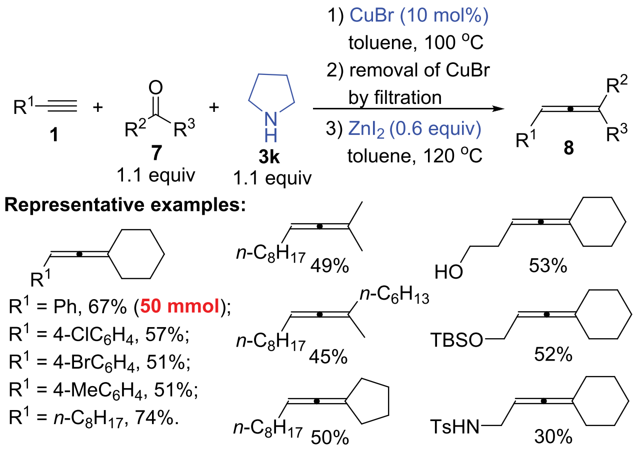



Since CuBr-catalyzed $KA^2$ reaction[22] and $ZnI_2$-promoted synthesis of trisubstituted allenes from tertiary propargylic amines[23] both proceeded smoothly in toluene, we tried to combine them in one pot and developed a practical three-step protocol to trisubstituted allenes **8** from terminal alkynes and ketones (Scheme 12).[23] The catalyst CuBr used in the first step must be removed by filtration; otherwise the reaction would give the trisubstituted allenes in much lower yields.

However, the substrate scope of ketones with the two methods described above was quite narrow: only cyclic ketones and methyl alkyl ketones worked well. Very recently, we have developed a direct and efficient one-pot approach for trisubstituted allenes **8** under the catalysis of CuI (10 mol %) with the help of $ZnBr_2$ (0.8 equiv) and $Ti(OEt)_4$ (2.0 equiv). Control experiments showed that $Ti(OEt)_4$ was indispensable for the first step of producing propargylic amine but did not participate in the second step; CuI was more efficient than $ZnBr_2$ in promoting the formation of propargylic amine, while $ZnBr_2$ was more effective in converting propargylic amine to allene than CuI. Notably,

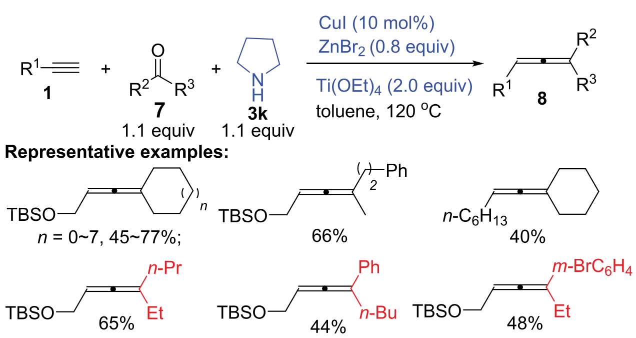

DOI: 10.1021/acs.accounts.9b00023
Acc. Chem. Res. 2019, 52, 1301−1312

**Scheme 12. Synthesis of Trisubstituted Allenes via CuBr/$ZnI_2$ Promoted Three-Step ATA Reactions from Terminal Alkynes, Ketones, and Pyrrolidine**




the scope of ketones has been extended to aryl alkyl ketones and dialkyl ketones (Scheme 13).[24]

**Scheme 13. Synthesis of Trisubstituted Allenes via CuI/$ZnBr_2$/$Ti(OEt)_4$ Promoted One-Pot ATA Reactions from Terminal Alkynes, Ketones, and Pyrrolidine**



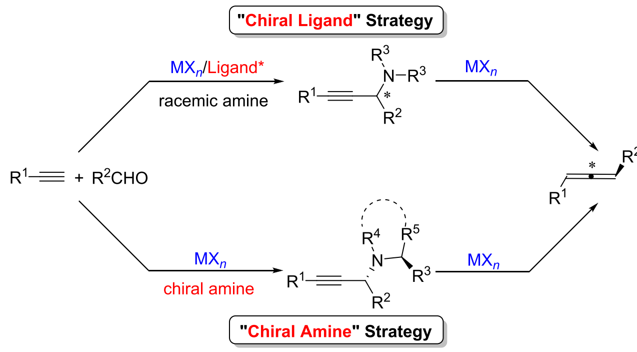

## 3. ENANTIOSELECTIVE ALLENATION OF TERMINAL ALKYNES (EATA) WITH ALDEHYDES

Based on the achievements of $ZnI_2$ or CuI-mediated racemic ATA reaction with aldehydes, we reasoned that there are two approaches for the easy synthesis of optically active 1,3-disubstituted allenes from terminal alkynes and aldehydes, the "chiral ligand" strategy and "chiral amine" strategy. The chiral ligand approach applies terminal alkynes, aldehydes, and nonchiral amine with the assistance of a proper chiral ligand, while the chiral amine approach utilizes terminal alkynes, aldehydes, and chiral amines (Scheme 14).[25] The key point of these two strategies is the formation of optically active propargylic amine in high ee, which is then transformed into the corresponding chiral 1,3-disubsituted allenes mediated by a matched metal salt under proper conditions.

### 3.1. Synthesis of Optically Active 1,3-Disubstituted Allenes via Chiral Ligand Strategy

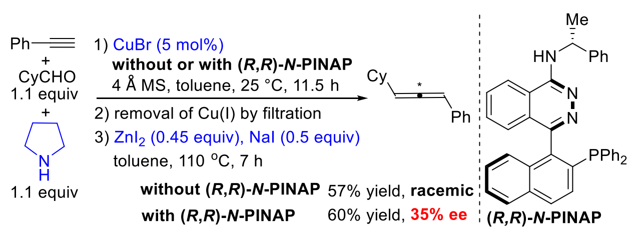
First, we developed a three-step approach for racemic 1-phenyl-3-cyclohexyl allene in toluene in 57% yield: (1) CuBr-catalyzed synthesis of propargyl amine from phenylacetylene, cyclohexanecarbaldehyde, and pyrrolidine at room temperature; (2) filtration to remove CuBr catalyst; (3) $ZnI_2$-mediated allene formation from the generated propargyl amine

**Scheme 14. Two Strategies for the Synthesis of Optically Active 1,3-Disubstituted Allenes**




in the presence of NaI at 110 °C. Inspired by the pioneering works by Knochel[26] and Carreira[27] for the slow formation of optically active propargylic amines, we considered to apply (*R*,*R*)-*N*-PINAP as the chiral ligand and pyrrolidine as the amine. The reaction proceeded efficiently and smoothly, affording the corresponding allene in 60% yield albeit with a rather low ee of 35% (Scheme 15).[25] Although very high enantioselectivities were observed for the preparation of chiral propargylic amines from the $A^3$ reaction with secondary-alkyl aldehydes,[27] a racemization occurred in the $ZnI_2$-promoted allene formation from the phenyl substituted propargylic amine.

**Scheme 15. Initial Attempt of "Chiral Ligand" Strategy**




We further reasoned that if a hydroxyl group was introduced at the $\alpha$-position of the C−C triple bond, the stereoselectivity in both steps may be improved by the possible coordination of the hydroxyl oxygen to the metal.[25] Excitingly, this hypothesis was working: the reaction of unprotected tertiary propargylic alcohols with different types of aldehydes including primary and secondary alkyl aldehydes, and benzaldehyde all afforded chiral 2,3-allenols in moderate yields with ≥90% ee. Moreover, when optically active secondary propargylic alcohols with a central chirality were applied, all four diastereoisomers could be conveniently prepared by this protocol with (*R*,*R*)-*N*-PINAP or (*R*,*S*)-*N*-PINAP as the ligand. However, primary propargylic alcohol did not work (Scheme 16).[25]

In 2015, we further developed a one-pot two-step protocol by replacing $ZnI_2$/NaI with $CdI_2$. Compared with the former three-step method, it was not necessary to remove CuBr by filtration (eq 1).[28] Prop-2-yn-1-ol and alkyl- or aryl-substituted simple terminal alkynes are still not compatible with this chiral ligand strategy.

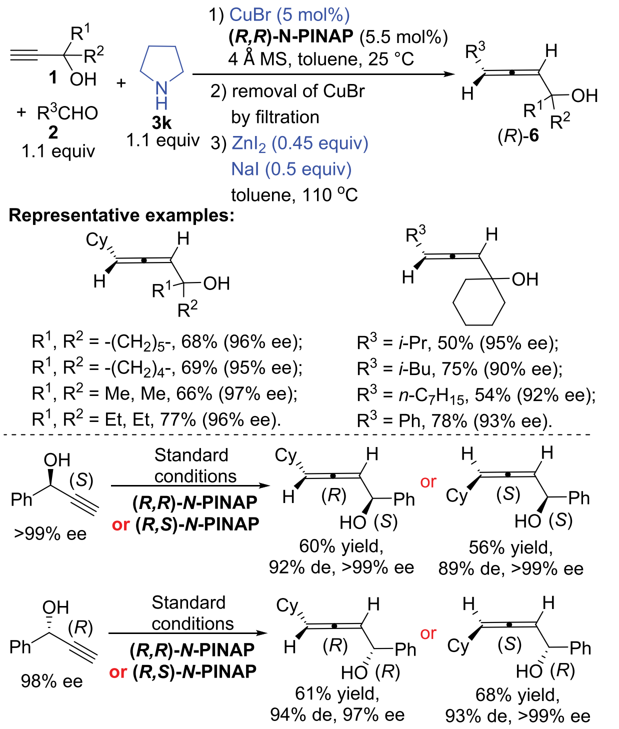
### 3.2. Synthesis of Optically Active 1,3-Disubstituted Allenes via Chiral Amine Strategy

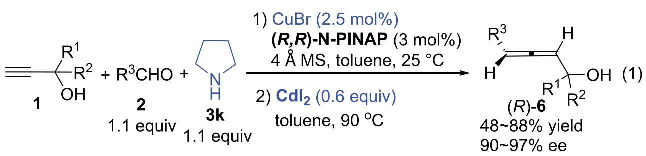
At the same time, we were also actively pursuing the chiral amine strategy. The initial attempt of chiral amine was based



DOI: 10.1021/acs.accounts.9b00023
Acc. Chem. Res. 2019, 52, 1301−1312

**Scheme 16. CuBr/$ZnI_2$-Mediated Three-Step Synthesis of Optically Active 1,3-Disubstituted α-Allenols via "Chiral Ligand" Strategy**

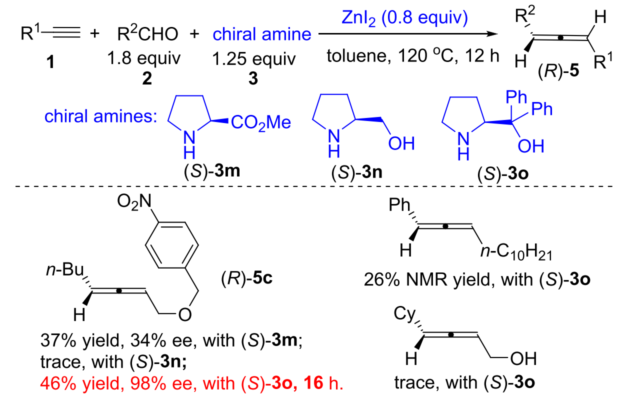




on the $KAuCl_4$- or $AgNO_3$-promoted incomplete synthesis of chiral 1,3-disubsituted allenes from L-prolinol-based propargylic amines reported by Che's group.[16] Thus, several proline derivatives (S)-**3m**−**3o** were first chosen as the resources of chiral amines for the $ZnI_2$-promoted EATA reaction with aldehydes. It was interesting to note that the diphenyl-substituted prolinol (S)-**3o** was the optimal chiral amine for the reaction of propargyl p-nitrobenzyl ether group with n-BuCHO in toluene, affording allene (R)-**5c** in 98% ee within 16 h. However, extensive studies showed that the scope of the reaction was extremely limited: simple terminal alkynes such as dodec-1-yne and prop-2-yn-1-ol both gave very poor results (Scheme 17).[25]

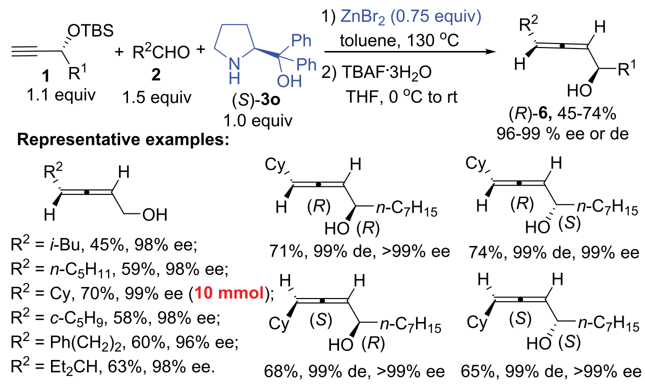
As we know, α-allenols with axial chirality are valuable building blocks in organic synthesis due to the synthetic potential of the hydroxyl group and the allene unit.[10] Although the chiral secondary and tertiary 2,3-allenols could be synthesized with the chiral ligand strategy,[25,28] the approach to chiral primary α-allenols is still not available. Inspired by the preparation of optically active allene (R)-**5c** bearing a sterically bulky benzyl ether group in excellent enantioselectivity, we envisioned that a TBS ether may serve this purpose due to the much stronger steric effect. To our delight, this idea was proven to be successful: $ZnBr_2$-promoted EATA reactions of TBS-protected prop-2-yn-1-ol with various aliphatic aldehydes afforded primary α-allenols (R)-**6** in 45−70% yields with 96−99% ee after deprotection of the TBS group. Notably, all of the

**Scheme 17. Initial Attempt of the "Chiral Amine" Strategy**



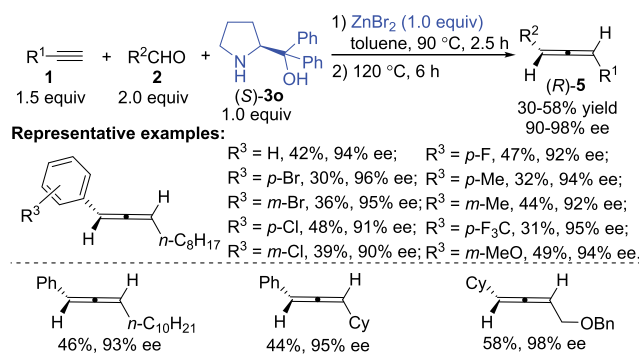
four diastereoisomers of secondary α-allenols could be highly stereoselectively prepared simply by adjusting the absolute configurations of the central chiralities in the TBS-protected secondary propargylic alcohols and α,α-diphenylprolinol (Scheme 18).[29] Unfortunately, this protocol is not applicable to aromatic and α,β-unsaturated aldehydes.

**Scheme 18. Synthesis of Chiral 1,3-Disubstituted α-Allenols via $ZnBr_2$-Promoted EATA Reaction with a TBS-Protection Strategy**



Later, a "two-stage" procedure for $ZnBr_2$-promoted EATA reactions of simple terminal alkynes with aromatic or aliphatic aldehydes was established to deliver 1,3-disubstituted allenes with 90−98% ee, although the yields are not very high (Scheme 19).[30]

**Scheme 19. Synthesis of Chiral Nonfunctionalized 1,3-Disubstituted Allenes via $ZnBr_2$ Promoted EATA Reaction**

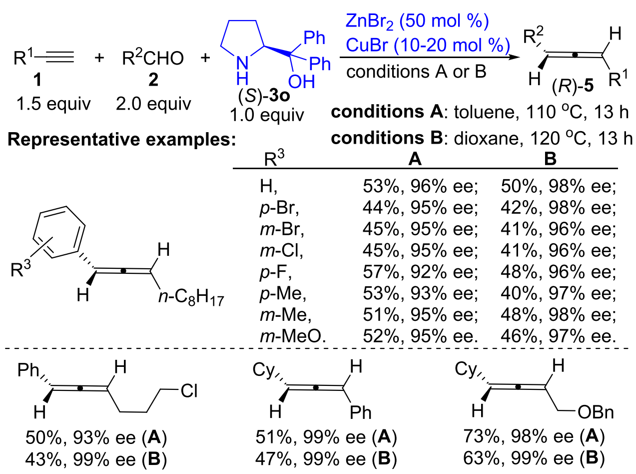




DOI: 10.1021/acs.accounts.9b00023
Acc. Chem. Res. 2019, 52, 1301−1312

Periasamy et al. also reported the synthesis of chiral 1,3-disubstituted allenes from simple nonfunctionalized terminal alkynes, aldehydes, and (*S*)-**3o**[31] based on our racemic version,[17] but we have not been able to reproduce their results in terms of ee and yield: the calibrated specific optical rotations of the same allenes with similar ee values from our study and the data in their report are different.[30] A combined approach using CuBr and $ZnBr_2$ was then developed, providing axially chiral 1,3-disubstituted allenes in somewhat higher yields with excellent enantioselectivity in toluene or dioxane (compare Scheme 20 with Scheme 19).[32] Control experiments revealed that CuBr was responsible for accelerating the formation of the propargylic amine intermediate while both CuBr and $ZnBr_2$ played crucial roles in the propargylic amine-to-allene transformation ($ZnBr_2$ was more efficient than CuBr in the second step).[32]

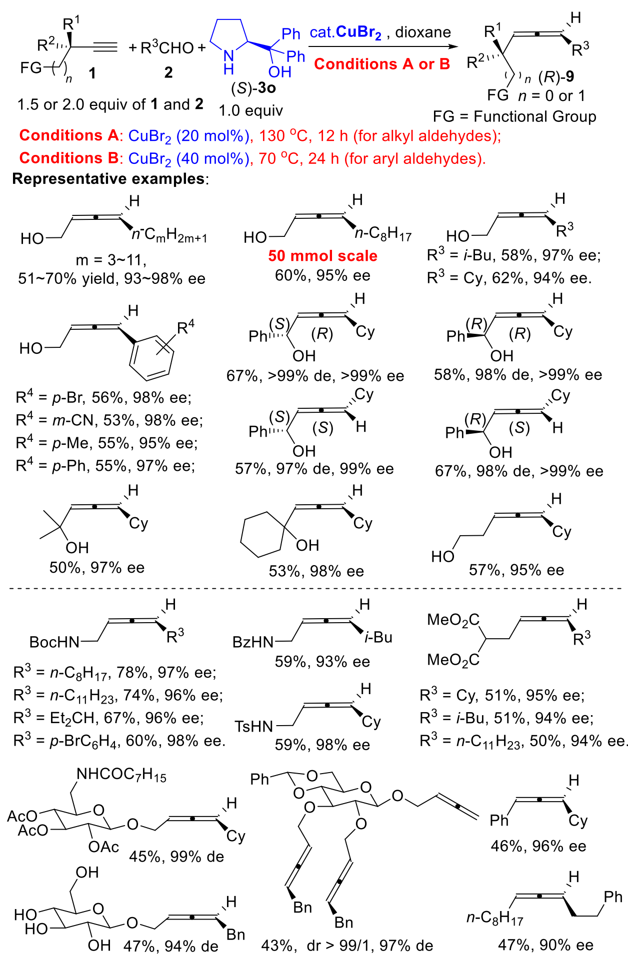
**Scheme 20. Synthesis of Chiral Nonfunctionalized 1,3-Disubstituted Allenes via CuBr/$ZnBr_2$-Copromoted EATA Reaction**




As mentioned above, chiral allenols are of great importance in organic synthesis. With the CuBr/$ZnI_2$- or CuBr/$CdI_2$-mediated multistep chiral ligand strategy, chiral tertiary and secondary $\alpha$-allenols could be prepared. On the other hand, chiral primary and secondary $\alpha$-allenols could be synthesized via the $ZnBr_2$-mediated chiral amine strategy from TBS-protected propargyl alcohols. However, there are several issues in common for these two strategies: (1) multistep manipulations are needed from propargylic alcohol; (2) aromatic aldehydes are not applicable; (3) chiral allenols bearing longer carbon chain could not be prepared; (4) nearly stoichiometric metal salt, $MX_n$, was used.

In 2015, we found by accident that $CuBr_2$ is the catalyst to address all those issues. Starting from unprotected terminal alkynols, various aldehydes, and (*S*)-**3o**, versatile 1,3-disubstituted chiral allenols were produced in moderate yields with excellent enantioselectivities. This method is quite general, tolerating primary, secondary, and tertiary terminal $\alpha$- and $\beta$-alkynols as well as alkyl and aryl aldehydes. The reaction may be easily scaled up to 50 mmol scale. For aryl aldehydes, the reaction should be conducted at 70 °C under the promotion of 40 mol % of $CuBr_2$. This newly developed $CuBr_2$-catalyzed EATA reaction was very impressive and can also be used for the preparation of different types of functionalized optically active 1,3-disubstituted allenes, such as 2,3-allenyl amides, 2,3-allenyl malonates, and carbohydrate-based 1,3-disubstituted allenes as well as nonfunctionalized chiral 1,3-disubstituted allenes such as 1,3-dialkyl allenes and 1-aryl-3-alkyl allenes(Scheme 21).[33]

**Scheme 21. Synthesis of Chiral 1,3-Disubstituted Allenes via $CuBr_2$-Catalyzed EATA Reaction without Protection**

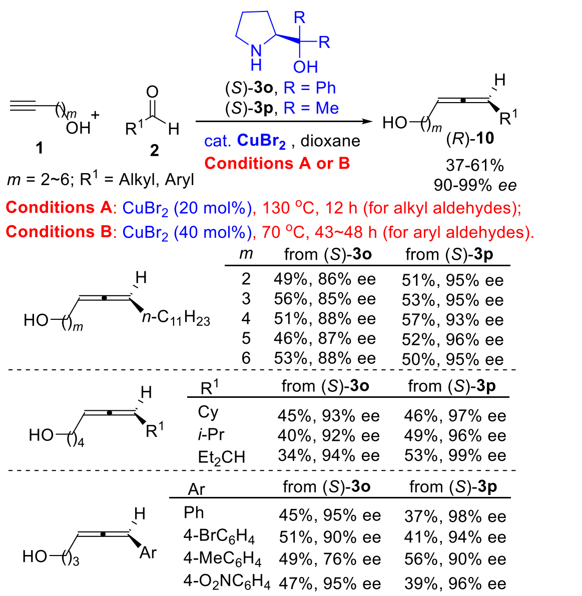



In the case of some chiral allenols with longer carbon chains (≥two carbon atoms) between the allene moiety and alcohol functionality, (*S*)-$\alpha,\alpha$-dimethylprolinol, (*S*)-**3p**, was used instead of (*S*)-$\alpha,\alpha$-diphenylprolinol, (*S*)-**3o** to obtain optically active 1,3-disubstuted allenols with higher ee-values (Scheme 22).[34] These allenols may be converted to the synthetically useful aldehyde, amide, bromide, mesylate, thiol, etc.[34]

The deuterium-labeling experiment applying a deuterated (*S*)-$\alpha,\alpha$-diphenylprolinol (*S*)-**3o-D$_2$** confirmed the 1,5-D transfer from the 5-position of (*S*)-**3o-D$_2$** to the 2-position of the allenol (*R*)-**6ad-D** with a $K_{H/D}$ of 2.0, indicating its rate-determining nature (Scheme 23A).[33a] The reaction between the in situ generated alkynylcopper species **Int7** and the iminium ion **Int8** via 1,2-attack of the alkynyl entity from the back-side of the dimethylhydroxymethyl or diphenylhydroxymethyl group would generate propargylic amine (*S*,*S*)-**Int9**, which underwent highly stereoselective $CuBr_2$-mediated intramolecular 1,5-hydride transfer followed by anti-$\beta$-elimination to deliver the (*R*)-allene unit. The reaction using (*S*)-dimethylprolinol may afford optically active propargylic


DOI: 10.1021/acs.accounts.9b00023
Acc. Chem. Res. 2019, 52, 1301−1312

**Scheme 22. Synthesis of Chiral 1,3-Disubstituted Allenols Bearing Long Carbon Chains via $CuBr_2$-Catalyzed EATA Reaction in the Presence of (*S*)-3o or (*S*)-3p**

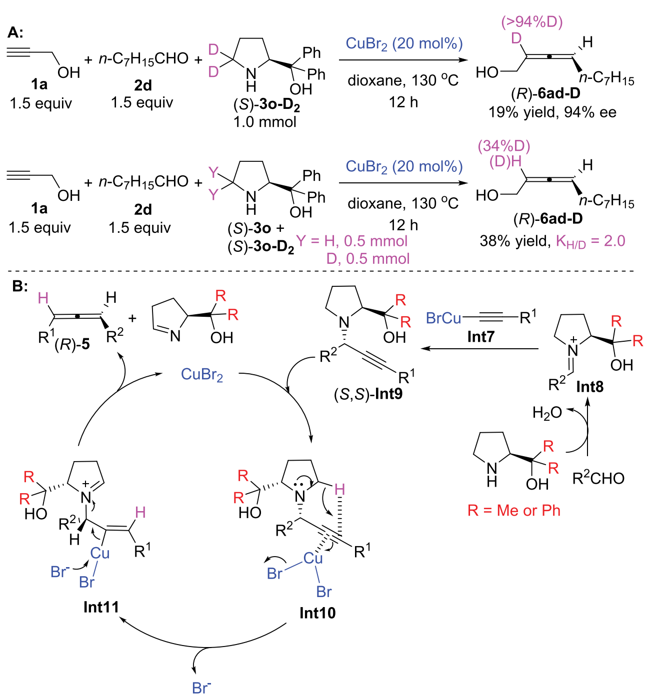



| m | from (S)-3o | from (S)-3p |
|---|---|---|
| 2 | 49%, 86% ee | 51%, 95% ee |
| 3 | 56%, 85% ee | 53%, 95% ee |
| 4 | 51%, 88% ee | 57%, 93% ee |
| 5 | 46%, 87% ee | 52%, 96% ee |
| 6 | 53%, 88% ee | 50%, 95% ee |

| $R^1$ | from (S)-3o | from (S)-3p |
|---|---|---|
| Cy | 45%, 93% ee | 46%, 97% ee |
| *i*-Pr | 40%, 92% ee | 49%, 96% ee |
| $Et_2CH$ | 34%, 94% ee | 53%, 99% ee |

| Ar | from (S)-3o | from (S)-3p |
|---|---|---|
| Ph | 45%, 95% ee | 37%, 98% ee |
| $4\text{-}BrC_6H_4$ | 51%, 90% ee | 41%, 94% ee |
| $4\text{-}MeC_6H_4$ | 49%, 76% ee | 56%, 90% ee |
| $4\text{-}O_2NC_6H_4$ | 47%, 95% ee | 39%, 96% ee |

**Scheme 23. (A) D-Labeling Study; (B) Proposed Mechanism**



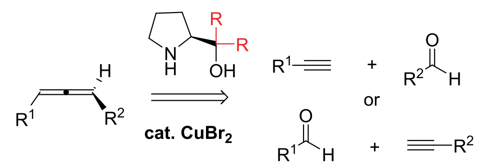

amine (*S*,*S*)-**Int9** in higher de, resulting in higher ee for 1,3-disubstituted allenes (Scheme 23B).[33,34]

## 4. APPLICATIONS OF EATA REACTION IN THE SYNTHESIS OF NATURAL PRODUCTS

### 4.1. Synthesis of Natural Products Containing Allene Moieties

Now, the synthesis of axially chiral 1,3-disubstituted allenes is so simple, just breaking them down to two different types of the corresponding aldehydes and terminal alkynes with most of the functionalities unprotected (Scheme 24).

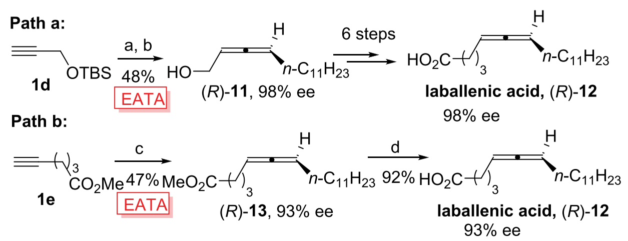
**Scheme 24. Retrosynthetic Analysis for Chiral 1,3-Disubstituted Allenes**




We then commenced to synthesize some of the typical 1,3-disubstituted allenic natural products, such as laballenic acid, insect pheromone, methyl (*R*)-8-hydroxyocta-5,6-dienoate, phlomic acid, and lamenallenic acid.[35]

**4.1.1. Synthesis of Laballenic Acid.**[36] Starting from TBS-protected propargyl alcohol **1d**, the EATA reaction with $n\text{-}C_{11}H_{23}CHO$ in the presence of (*S*)-**3o** afforded optically active *α*-allenol (*R*)-**11** in 48% yield with 98% ee after deprotection. After the transformation to 2,3-allenyl malonate, subsequent decarboxylation, reduction with DIBAL-H, Wittig olefination, and oxidation with $Ag_2O$ afforded the final product laballenic acid in 12% overall yield with 98% ee by starting from **1d** (Scheme 25, path a).[37] A much more efficient route for the synthesis of laballenic acid in 93% ee has also been developed by applying the $CuBr_2$-catalyzed EATA reaction from readily available methyl 5-hexynoate **1e** in just two steps (Scheme 25, path b).[33b]

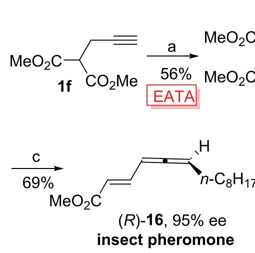
**Scheme 25. Synthesis of Laballenic Acid**




**Reagents and conditions:**
**Path a:** a) $ZnBr_2$ (0.75 equiv), **1d** (1.1 equiv), $n\text{-}C_{11}H_{23}CHO$ (1.5 equiv), (*S*)-**3o** (1.0 equiv), toluene, 130 °C, 12 h; b) TBAF, THF, 0 °C to rt, 2 h
**Path b:** c) $CuBr_2$ (20 mol%), **1e** (1.5 equiv), $n\text{-}C_{11}H_{23}CHO$ (1.5 equiv), (*S*)-**3o** (1.0 equiv), dioxane, 130 °C, 12 h; d) KOH (2.5 equiv), $MeOH/H_2O$ = 4/1, 60 °C, 2 h.

**4.1.2. Synthesis of Insect Pheromone (*R*)-16.**[38] Starting from dimethyl 2-(prop-2-yn-1-yl)malonate **1f**, we synthesized insect pheromone (*R*)-**16** in 95% ee with three steps based on the $CuBr_2$-catalyzed EATA reaction of 2-propargyl malonate **1f**, octanal, and (*S*)-**3o** (Scheme 26).[33b]

**Scheme 26. Synthesis of (*R*)-16**

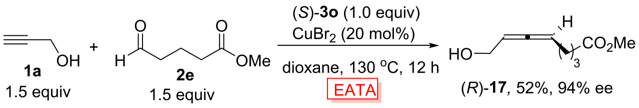



**Reagents and conditions:** a) $CuBr_2$ (20 mol%), **1f** (2 equiv), $n\text{-}C_8H_{17}CHO$ (2 equiv), (*S*)-**3o** (1 equiv), dioxane, 130 °C, 12 h; b) LiCl (2 equiv), $H_2O$ (4 equiv), DMSO, 180 °C, 1.5 h; c) (1) LDA (2.2 equiv) in THF, -78 °C; (2) PhSeSePh (1.2 equiv); (3) $NaIO_4$ in $THF/H_2O$.


DOI: 10.1021/acs.accounts.9b00023
Acc. Chem. Res. 2019, 52, 1301−1312

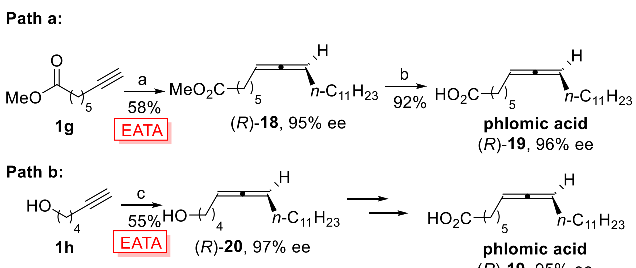
**4.1.3. Synthesis of Methyl (*R*)-8-Hydroxyocta-5,6-dienoate.**[39] The beauty of functionality tolerance of the $CuBr_2$-catalyzed EATA reaction was further demonstrated by the just one-step enantioselective synthesis of (*R*)-8-hydroxyocta-5,6-dienoate, (*R*)-**17**, in 94% ee from propargyl alcohol **1a**, 5-methoxycarbonylbutanal **2e**, and (*S*)-**3o** (Scheme 27).[33b]

**Scheme 27. Synthesis of (*R*)-17**




**4.1.4. Synthesis of Phlomic Acid.** The $CuBr_2$-catalyzed EATA reaction of **1g**, $n$-$C_{11}H_{23}CHO$, and (*S*)-**3p** produced (*R*)-**18** in 58% yield with 95% ee. After hydrolysis, phlomic acid was synthesized for the first time (Scheme 28, path a).[40]

**Scheme 28. Syntheses of (*R*)-19**

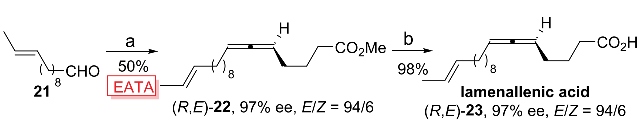



Reagents and conditions:
Path a: a) **1g** (1.5 equiv), $n$-$C_{11}H_{23}CHO$ (1.5 equiv), (*S*)-**3p** (1.0 equiv), $CuBr_2$ (20 mol%), dioxane, 130 °C, 12 h; b) KOH (2.5 equiv), $MeOH/H_2O$ = 4/1, 60 °C, 3 h.
Path b: c) **1h** (1.0 equiv), $n$-$C_{11}H_{23}CHO$ (1.4 equiv), (*S*)-**3p** (1.2 equiv), $CuBr_2$ (20 mol%), dioxane, 130 °C, 12 h.

Recently, an alternative synthesis starting from hex-5-yn-1-ol **1h** has also been developed. The $CuBr_2$-catalyzed EATA reaction of terminal alkyne **1h**, $n$-$C_{11}H_{23}CHO$, and (*S*)-**3p** gave chiral 5-allenol (*R*)-**20** in 55% yield with 97% ee. The treatment of (*R*)-**20** with $PPh_3$, imidazole, and $I_2$ yielded an iodide, which was transformed to the target phlomic acid in 95% ee in a couple steps (Scheme 28, path b).[34] It should be noted that (*S*)-$\alpha,\alpha$-dimethylprolinol (*S*)-**3p** was applied instead of (*S*)-$\alpha,\alpha$-diphenylprolinol (*S*)-**3o** in the EATA reaction to get higher enantioselectivity in both routes.

**4.1.5. Synthesis of Lamenallenic Acid.**[41] The $CuBr_2$-catalyzed EATA reaction of **1e** with *trans* 10-dodecenal **21** in the presence of (*S*)-**3p** afforded the methyl ester of lamenallenic acid (*R*,E)-**22** in 50% yield with 97% ee and an *E*/*Z* ratio of 94/6. After hydrolysis, the lamenallenic acid was delivered with 97% ee and an *E*/*Z* selectivity of 94/6 (Scheme 29).[42,43]

**Scheme 29. Synthesis of Lamenallenic Acid**



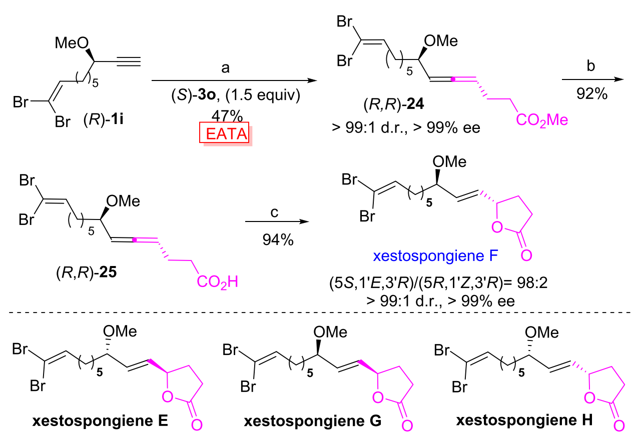

Reagents and conditions: a) **21** (1.1 equiv), **1e** (1.1 equiv), (*S*)-**3p** (1.0 equiv), $CuBr_2$ (20 mol%), dioxane, 130 °C, 12 h; b) KOH (2.5 equiv), $MeOH/H_2O$ = 4/1, 60 °C, 3 h.

## 4.2. Applications of EATA Reactions in the Syntheses of Natural $\gamma$-Butyrolactones

As we know, $\gamma$-(*E*)-alkenylic and $\gamma$-alkylic $\gamma$-butyrolactones are two different types of lactones with interesting biological activities that exist extensively in animals and plants.[44] Thus, much effort toward them has been made by chemists. However, direct and efficient access to $\gamma$-(*E*)-vinylic $\gamma$-butyrolactones, especially the asymmetric version, is still extremely limited. Very recently, we have developed a modular AuCl(LB-Phos)-catalyzed efficient and stereoselective cycloisomerization of allene approach to (*E*)-vinylic $\gamma$-butyrolactones from optically active allenoic acids constructed via the $CuBr_2$-catalyzed EATA reactions with aldehydes.[45]

**4.2.1. Synthesis of Xestospongienes E, F, G, and H.** Based on this cycloisomerization strategy, the first synthesis of racemic xestospongiene as well as the enantioselective syntheses of xestospongienes E, F, G, and H have been realized.[45] Notably, the absolute configurations of the chiral centers in xestospongienes E and F have been revised. Take the synthesis of xestospongiene F as an example: The $CuBr_2$-catalyzed EATA reaction of (*R*)-**1i**, methyl 4-oxobutanoate, and (*S*)-**3o** in a ratio of 1:1.5:1.5 afforded allene (*R*,*R*)-**24** as a single stereoisomer in 47% yield with >99/1 dr and >99% ee. Hydrolysis of (*R*,*R*)-**24** was conducted subsequently by its treatment with $LiOH \cdot H_2O$ at 90 °C for 1.5 h affording (*R*,*R*)-**25**, which was cycloisomerized under the catalysis of 10 mol % of AuCl(LB-Phos)[46] at −30 °C to afford xestospongiene F (reported as xestospongiene E[44a]) in 94% yield with >99/1 dr and >99% ee (Scheme 30). The remaining three optical isomers could also be obtained easily in a similar way by just replacing amino alcohol (*S*)-**3o** with (*R*)-**3o** or propargylic alcohol (*R*)-**1i** with (*S*)-**1i**.

**Scheme 30. Synthesis of Xestospongienes E, F, G, and H**




Reagents and conditions: a) methyl 4-oxobutanoate (1.5 equiv), $CuBr_2$ (20 mol%), dioxane, 120 °C; b) $LiOH \cdot H_2O$ (1.5 equiv.), $EtOH/H_2O$ = 1:1, 90 °C; c) AuCl(LB-Phos) (10 mol%), AgOTs (10 mol%), $CHCl_3$, -30 °C.

**4.2.2. Synthesis of (*R*)-4-Tetradecalactone, (*S*)-4-Tetradecalactone, (*R*)-$\gamma$-Palmitolactone, and (*R*)-4-Decalactone.** In addition, the syntheses of naturally occurring $\gamma$-alkylic $\gamma$-lactones such as (*R*)-4-tetradecalactone, (*S*)-4-tetradecalactone, (*R*)-$\gamma$-palmitolactone, and (*R*)-4-decalactone have also been achieved by applying a C−O bond cleavage-free hydrogenation.[45] Taking the synthesis of (*R*)-4-tetradecalactone as an example, the $CuBr_2$-catalyzed EATA reaction of ethyl pent-4-ynoate **1j**, nonanal, and (*S*)-**3o** afforded (*R*)-**26**, which was converted to chiral allenoic acid (*R*)-**27** in 39% yield


DOI: 10.1021/acs.accounts.9b00023
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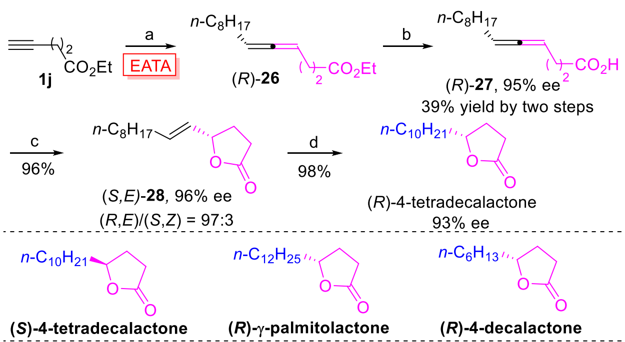
with 95% ee. Subsequent AuCl(LB-Phos)-catalyzed cycloisomerization of (*R*)-**27** was executed to yield $\gamma$-butyrolactone bearing a *trans* C=C bond, (*S,E*)-**28**, in 96% yield with 96% ee and 97:3 *E*/*Z* ratio. (*R*)-4-Tetradecalactone was obtained by hydrogenation of (*S,E*)-**28** under the catalysis of Pd/C (5 mol %) in 98% yield with 93% ee (Scheme 31). (*S*)-4-Tetradecalactone was prepared according to this route by replacing (*S*)-**3o** with (*R*)-**3o**. (*R*)-$\gamma$-Palmitolactone and (*R*)-4-decalactone were prepared according to this route by replacing nonanal with the corresponding aldehydes.

**Scheme 31. Synthesis of (*R*)-4-Tetradecalactone, (*S*)-4-Tetradecalactone, (*R*)-$\gamma$-Palmitolactone, and (*R*)-4-Decalactone**




**Reagents and conditions:** a) **1j** (1.2 equiv), *n*-$C_8H_{17}$CHO (1.5 equiv), (*S*)-**3o** (1.0 equiv), $CuBr_2$ (20 mol%), dioxane, 120 °C, 16 h; b) $LiOH{\cdot}H_2O$ (1.5 equiv), EtOH/$H_2O$ = 1:1, 90 °C, 18 h; c) AuCl(LB-Phos) (5 mol%), AgOTs (5 mol%), $CHCl_3$, 25 °C, 3 h; d) Pd/C (5 mol%), $H_2$ (25 atm), EtOAc, rt, 36 h.

## 5. CONCLUSIONS AND PERSPECTIVES

In this Account, we have summarized our recent studies on the allenation of terminal alkynes (ATA) with paraformaldehyde, aromatic and aliphatic aldehydes, and ketones in the presence of a proper amine to afford monosubstituted allenes, 1,3-disubstitued allenes, and trisubstituted allenes, respectively. On the other hand, we have also demonstrated chiral ligand strategy and chiral amine strategy for the enantioselective allenation of terminal alkynes (EATA) with aldehydes to produce optically active 1,3-disubstituted allenes. Notably, the EATA reactions have been applied in the efficient and highly enantioselective syntheses of some natural 1,3-disubstituted allenes including laballenic acid, insect pheromone, methyl (*R*)-8-hydroxyocta-5,6-dienoate, phlomic acid, and lamenallenic acid, as well as some non-allene natural $\gamma$-butyrolactones including xestospongienes (E, F, G, and H), (*R*)-4-tetradecalactone, (*S*)-4-tetradecalactone, (R)-$\gamma$-palmitolactone, and (*R*)-4-decalactone. Owing to the readily available starting materials and easy operations, the ATA reactions will stimulate the research on allene chemistry with no doubt. The achievements of EATA reactions are promising and provide new opportunities in constructing central chiralities via axis-to-center approach.

Of course, there are still some unconquered challenges: (1) Since the reactions are usually performed above reflux conditions, it would be better if the reaction may be conducted under milder conditions for some sensitive functionalities with better catalytic recipes. (2) The EATA reaction via the chiral ligand strategy is quite underdeveloped. The scope of terminal alkynes is limited to tertiary and secondary $\alpha$-alkynols. New ligands should be designed to upgrade this technology. (3) The EATA reaction with ketones has not been realized. We are actively working to overcome these challenges and apply these developed methodologies in target synthesis.

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### Notes

The authors declare no competing financial interest.

### Biographies

**Xin Huang** was born in 1987 in Jiangsu, China. He received his B.S. degree in chemistry from Nanjing University (2009) and his Ph.D. degree from Zhejiang University (2014) under the supervision of Professor Shengming Ma. His doctoral research was focused on the synthesis and cyclization reactions of allenes. He left Prof. Ma's group in January 2016 and joined Chinese Peptide Company as a senior process researcher until August 2017. Then he came back to Prof. Ma's group as a postdoctoral researcher to pursue an academic career. His current research interests focus on the syntheses and applications of phosphorous ligands and the radical reaction of allenes.

**Shengming Ma** was born in 1965 in Zhejiang, China. He received his Ph.D. from Shanghai Institute of Organic Chemistry (SIOC) and became an assistant professor there in 1991. After postdoctoral research at the ETH with Prof. Venanzi and Purdue University with Prof. Negishi, he returned to SIOC in 1997. From February 2003 to September 2007, he was jointly appointed by SIOC and Zhejiang University (ZJU). In October 2007, he moved to East China Normal University to help build the research program in organic chemistry. After finishing the duty at ECNU, starting from 2014, he is a professor at Fudan University. At the same time, he is also a research professor at SIOC, Qiu Shi at ZJU, and Adjunct Professor at CUHK. He received the Mr. & Mrs. Sun Chan Memorial Award in Organic Chemistry (2004), OMCOS Springer Award (2005), National Award for Research in Natural Science in China (Second-Class, 2006), and Natural Science Award of Shanghai (First-Class, 2010).

## ■ ACKNOWLEDGMENTS

We gratefully acknowledge all the co-workers that have participated for their significant contributions to the projects described herein. Financial support from National Natural Science Foundation of China (Grant No. 21690063) and National Basic Research Program of China (2015CB856600) is greatly appreciated.

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DOI: 10.1021/acs.accounts.9b00023
*Acc. Chem. Res.* 2019, 52, 1301−1312